

Journal of the Air & Waste Management Association

ISSN: 1096-2247 (Print) 2162-2906 (Online) Journal homepage: www.tandfonline.com/journals/uawm20

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To cite this article: Kunning Zhu, Junfeng (Jim) Zhang & Paul J. Lioy (2007) Evaluation and Comparison of Continuous Fine Particulate Matter Monitors for Measurement of Ambient Aerosols, Journal of the Air & Waste Management Association, 57:12, 1499-1506, DOI: 10.3155/1047-3289.57.12.1499

To link to this article: https://doi.org/10.3155/1047-3289.57.12.1499

Published online: 24 Jan 2012.

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TECHNICAL PAPER

ISSN:1047-3289 *J. Air & Waste Manage. Assoc.* **57**:1499–1506
DOI:10.3155/1047-3289.57.12.1499
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Evaluation and Comparison of Continuous Fine Particulate Matter Monitors for Measurement of Ambient Aerosols

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ABSTRACT

To provide a scientific basis for the selection and use of continuous monitors for exposure and/or health effects studies, and for compliance and episode measurements at strategic locations in the State of New Jersey, we evaluated the performance of seven continuous fine particulate matter ($PM_{2.5}$) monitors in the present study. Gravimetric samplers, as reference methods, were collocated with real-time instruments in both laboratory and field tests. The results of intercomparison of real-time monitors showed that the two nephelometers used in this study correlated extremely well ($r^2 \sim 0.97$), and two tapered element oscillating monitors (TEOM 1400 and TEOM filter dynamics measurement system [FDMS]) correlated well ($r^2 > 0.85$), whereas two beta gauges displayed a weaker correlation ($r^2 < 0.6$). During a summertime controlled (laboratory) evaluation, the measurements made with the gravimetric method correlated well with the 24-hr integrated measurements made with the real-time monitors. The SidePak nephelometer overestimated the particle concentration by a factor of approximately 3.4 compared with the gravimetric method. During a summertime field evaluation, the TEOM FDMS monitor reported approximately 30% higher mass concentration than the Federal Reference Method (FRM); and the difference could be explained by the loss of semi-volatile materials from the FRM sampler. Results also demonstrated that 24-hr average $PM_{2.5}$ mass concentrations measured by beta gauges and TEOM (50 °C) in winter correlated well with the integrated gravimetric method. Seasonal differences were observed in the performance of the TEOM (50 °C) monitor in measuring the particle mass attributed to the higher semi-volatile material loss in the winter weather. In applying the real-time particulate matter monitoring data into Air Quality Index (AQI) reporting, the Conroy method and the 8-hr end-hour average method were both found to be suitable.

IMPLICATIONS

To provide information on continuous monitoring is to help examine the attainment of the National Ambient Air Quality Standards $PM_{2.5}$ standard. Because the gravimetric method demands a large amount of effort and resources, its replacement of the with real-time instruments could reduce the costs of sampling and provide equivalent or better data quality for use in attainment analyses. Real-time data can also be used for more timely alert and warning of severe particulate air pollution. The application of continuous monitors was addressed in the U.S. Environmental Protection Agency criteria document of $PM_{2.5}$, and present research provides important supporting information on the utility of these monitors.

INTRODUCTION

Numerous epidemiological studies have yielded results about human exposure to particulate matter (PM) and associated health responses. These have shown strong associations between the elevated levels of PM and increases in mortality and morbidity (e.g., exacerbation of asthma and other respiratory diseases, lung function decrements, and cardiovascular disease).[1–7] Fine particles smaller than 2.5 μm in aerodynamic diameter ($PM_{2.5}$) are of the greatest concern owing to their size and transportability in the human body. The U.S. Environmental Protection Agency (EPA) estimates that tens of thousands of premature deaths yearly are associated with exposure to excess levels of $PM_{2.5}$.[8] On the basis of the conclusions of many studies, the new National Ambient Air Quality Standards (NAAQS) $PM_{2.5}$ standard was promulgated by EPA in 1997 and upheld by the U.S. Supreme Court in 2002.[9] However, the current NAAQS $PM_{2.5}$ standard only regulates to achieve 24-hr and annual average mass concentrations. These do not address transient elevations in fine particle concentrations. Several studies have found that acute exposure to particulate air pollution may trigger or exacerbate cardiovascular events such as myocardial ischemia, peripheral thrombosis, acute arterial vasoconstriction, and progression of atherosclerosis in

sensitive populations.[10,11] In addition to having the advantage of real-time monitoring, continuous instruments may have potentially higher accuracy than a discrete filter-based gravimetric method due to a decreased loss of semi-volatile material.[12] For example, the Federal Reference Method (FRM) in Houston, TX and Seattle, WA was observed to underestimate ambient PM mass when compared with continuous methods.[13]

The goal of this paper is to provide an evaluation and recommendations associated with the operation of six continuous $PM_{2.5}$ monitors under a variety of conditions. This evaluation will help improve the regulatory compliance measurements using these devices. Ambient aerosols of New Jersey were used for testing in a summer and winter season in a controlled environment facility (CEF) and in real-world settings, respectively.

EXPERIMENTAL METHODS

$PM_{2.5}$ Monitors under Evaluation

The continuous monitors being evaluated were selected on an as-available basis, i.e., the models that have been commonly used for research and compliance purposes and that were available to us. The instrument models and their operating principles are summarized in Table 1. The nephelometer, SidePak, was factory calibrated against Arizona road dust.

Evaluation in the CEF

The CEF used was a 25-m^3 stainless steel chamber in which temperature, humidity, and air-exchange rate were controlled, and the air supplied was treated by a series of conditioning processes including cooling/heating and humidification/dehumidification processes. Activated carbon and high efficiency particulate arrestance (HEPA) filters were used to remove ambient PM when testing the effects of relative humidity (RH) on the monitor performances. Eight small brushless fans were installed in the CEF to completely mix the chamber.

To test the relationship between a continuous monitor and the Harvard Impactor (HI) gravimetric method, the CEF incoming air filters (HEPA and activated carbon) were removed from the pathway of sampling air to the CEF before another set of experiments that sampled ambient air. Six continuous particle mass monitors and a HI sampler were collocated in the CEF. The two sampling campaigns were conducted from January to March 2004, and from July to August 2004, respectively. The temperature and RH of the sampled outdoor air entering the CEF were monitored by the beta gauge (Metone) and validated by the sensors installed within the CEF.

The HI sampler used 37-mm diameter Teflon filters that were conditioned in the weighing room at a temperature of 20 °C and RH of approximately 30–40% and weighed three times before and after sampling. A MT5 microbalance (Mettler Toldeo Inc.) was used for weighing the filters. The balance had a sensitivity of ± 1 μg. The weighing procedures followed guidelines in EPA's Quality Assurance Program.[14] The tolerance for triplicate weighing of filters was in agreement within ± 3 μg. The mean net mass of the field blank and weighing room blank were examined and observed to have minimal changes (less than ± 5 μg), thus, they were not subtracted from the net mass of sample filters. The sample volume was calculated as the product of the sample duration and the mean of pre- and post sampling flow rates measured by an electronic flow meter (DryCal DC-Lite, Bios Inc.). The operation of continuous monitors strictly followed the manufacturer's manuals.

Evaluation in the Field

The field sampling site (40°17′ N and 74°49′ W, elevation 59 m) was located at the Technical Center of the New Jersey Department of Environmental Protection, Ewing, NJ. The trailer used for sampling was set up at a street with a perpendicular distance of 100 m from a heavily trafficked freeway. The site also lies 500 m north of the Trenton Mercer Airport that accommodates patrol jets. Parking lots, office buildings, roads, and woods surrounded the site. Construction work was being completed in an office building about 50 m away from the trailer

Table 1. Continuous PM monitors used in this study.

	β-Radiation Attenuation Monitor	TEOM	Light Scattering Monitor
Principle	The intensity of the β radiation is proportional to the particle mass.	The frequency of a vibrating oscillator is proportional to the particle mass.	The intensity of the scattered light is proportional to particle mass.
Quantification	$x = \frac{1}{\mu} \ln \frac{I_0}{I}$	$Dm = K\left(\frac{1}{f_v^2} - \frac{1}{f_a^2}\right)$	$b_{sp} = C\lambda^{-a}$
	where x = mass density of the absorbing matter, μ = absorption cross-section of the material, and I_0, I = measured β-ray intensity of clean and dust-laden filter tape, respectively.	where Dm = change of mass on the oscillating filter, K = spring constant, and f_v and f_a = oscillation frequency after and before the test, respectively.	where b_{sp} = light scattering coefficient, C = aerosol concentration, λ = wavelength, and α = Angstrom exponent.
Manufacturer	Metone, Thermal	Thermal (Formerly R&P Inc.)	TSI, Radiance
Sampling line	Metone was heated to 50 °C throughout the study, whereas Thermal was heated only during winter;	TEOM (50C) was heated to 50 °C, whereas FDMS was heated to 30 °C	Instruments were not implemented with heating.
Inlet	R&P PM_{10} and R&P sharp cut cyclone inlets	R&P PM_{10} and R&P sharp cut cyclone inlets	TSI $PM_{2.5}$ Impactor[a]
Flow rate	16.67 l/min	16.67 l/min	1.7 l/min

Notes: [a]No size selective inlet used for radiance nepholometer.

during two sampling periods. The monitoring data we collected during September 2004 and December 2004 to February 2005.

The tapered element oscillating monitors filter dynamics measurement system (TEOM FDMS), TEOM (50C), beta gauge (Anderson), and beta gauge (Metone) were collocated inside the trailer with their sample inlets located approximately 1 m above the trailer roof. The FRM (Partisol Model 2025, R&P) was located outside the trailer at a distance of 3 m. The major difference between the field tests in Ewing and the CEF test was that the ambient aerosol was directly drawn from the ambient environment into the continuous monitors during the field test, whereas the air sampled for the CEF test was conditioned for specific variables (e.g., RH). Another difference was that a manufacturer (Anderson Inc.) provided heating devices for the beta gauge (Anderson) used in the winter field campaign.

The distance between two inlets of any continuous monitors was greater than 1 m to avoid potential interferences. A solar powered weather station (Davis Inc.) was also located on the trailer roof. Inside the trailer, temperature was controlled at a range of 0–30 °C to maintain suitable operation conditions for the electronic units in real-time monitors. For practical considerations, the SidePak and Radiance nephelometers were not tested during this comparison because they are designed for personal monitoring and did not have weather-protected inlets. The filter samples were transported in a cooler.

Air Quality Index Reporting

The Air Quality Index (AQI) has been developed to facilitate the public's understanding of air quality. Converting actual $PM_{2.5}$ concentrations to AQI values allows an easier understanding by the general public that an AQI value of less than 100 is "okay" and a value of 100 or greater represents various degrees of "bad air quality". The current controlling $PM_{2.5}$ air quality standard is based on a 24-hr averaged value or an annual average. However, with a real-time monitor it is possible to provide a more rapid measurement of ambient $PM_{2.5}$ concentrations and the variability of these concentrations. Several methods, including the end-hour average method, 24-hr mid-average method, and the Conroy Method,[15] have been proposed to report the AQI using the real-time measurements of $PM_{2.5}$ as described in Table 2. In the end-hour average method, every hourly average is treated equally. In the Conroy method, the most recent 4-hr average is weighed more heavily and causes the computed concentration to be responsive to the hourly changes. In the same manner, the 24-hr mid-average method most weighed toward the current hourly average.

RESULTS AND DISCUSSION

Continuous $PM_{2.5}$ Measurements in the CEF

Maximum values, mean concentrations, and SD for contemporaneous hourly $PM_{2.5}$ mass measurements for the CEF campaigns are listed in Table 3. Short-term negative values were observed at low ambient particle levels (<5 $\mu g/m^3$) and could be a result of incorrect zero background coefficients or due to significant desorption or volatilization of PM components from the filters.

Table 2. Methods used to calculate concentration for air quality index prediction.

Method	Description
End-hour average method	$\frac{\sum_{t=0}^{h-1} X_{h-t}}{h}$
24-hr mid-average method	$\frac{X_0 + \frac{\sum_{t=0}^{-10} X}{11} + \frac{\sum_{t=1}^{12} X}{12}}{3}$
Conroy Method[15]	$\frac{\frac{\sum_{t=0}^{-11} X}{12} + \frac{\sum_{t=0}^{-3} X}{4}}{2}$

Notes: X_t represents the hourly average concentration at the t hour, the positive t represents the t hour in the future and negative t represents the t hour in the past.

CEF Winter Study. A total of 164 hourly contemporaneous samples were obtained with each method and used in the comparison. The agreement among the measurements obtained for five continuous samplers was examined using a one-way analysis of variance (ANOVA). On the basis of the estimated 95% confidence intervals, no significant difference was observed between most paired continuous measurements. Two exceptions were the TEOM (50C) monitor and beta gauge (Metone).

CEF Summer Study. A total of 185 contemporaneous samples were obtained from each method and used for intersampler concentration comparisons. A one-way ANOVA found no significant difference among all contemporaneous measurements during the summer campaign (see Table 3). The result suggests that all five continuous monitors had similar performance in measuring the variation of $PM_{2.5}$ mass under summer conditions.

Table 4 lists the results from a regression analysis using the measurements obtained during the summer study. Strong correlations were observed for all paired gravimetric and continuous monitors. Compared with the gravimetric method, the regression slopes for the TEOM (FDMS), beta gauge (Anderson), and SidePak nephelometer were significantly higher than 1, indicating that a consistently higher concentration was measured by these real-time monitors. The regression slope for the TEOM (50C) monitor was less than 1, indicating an underestimation of the mass. The beta gauge (Metone) monitor performed similarly to the gravimetric method in the measurement of particle mass and had a regression slope of 1.01 ± 0.05.

When the ambient $PM_{2.5}$ level was less than 5 $\mu g/m^3$, the TEOM (50C) estimated a particle mass that was greater than that reported by the HI sampler. This artifact of the TEOM (50C) measurement is due to a factory setting that added 3 $\mu g/m^3$ to the actual reading that, when subtracted, would yield similar results. Table 4 showed that

Table 3. Maximum and mean concentrations ($\mu g/m^3$) of contemporaneous $PM_{2.5}$ mass during sampling campaign in CEF:[1] TEOM (50C) located at the DEP site in New Brunswick, NJ.

	Winter (January 1 to March 7, 2004)		Summer (July 10 to August 8, 2004)	
Sampler	Maximum ($\mu g/m^3$)	Average ± SD ($\mu g/m^3$)	Maximum ($\mu g/m^3$)	Average ± SD ($\mu g/m^3$)
TEOM (50C)[1]	26.0	10.5 ± 5.9	37.0	10.7 ± 8.8
TEOM (50C)	20.2	9.3 ± 4.5	31.0	8.1 ± 7.0
TEOM (FDMS)	28.9	10.0 ± 7.9	35.2	9.7 ± 8.4
Beta gauge (Metone)	39.0	12.5 ± 9.6	38.0	10.3 ± 8.6
Beta gauge (Anderson)	32.5	9.6 ± 8.0	42.5	9.3 ± 10.2
SidePak nephelometer	71.3	25.4 ± 23.2	93.2	22.2 ± 24.5

the beta gauge (Anderson) and the TEOM (FDMS) reported approximately 25 and 15%, respectively, higher mass than the HI sampler during the summer. The higher values measured by these two continuous monitors mainly occurred when the ambient PM level was greater than 20 $\mu g/m^3$.

Continuous $PM_{2.5}$ Measurements at the Field Site

Field Winter Study. The measurements made by all the real-time monitors and the FRM during the winter campaign are illustrated in Figure 1. For comparisons with the FRM, the hourly $PM_{2.5}$ data of continuous monitors were averaged over 24 hr. The TEOM (FDMS) monitor was identified as having a technical problem on the tenth sampling day, and measurements after that date were not included in the data analysis. However, using the same sampling days for comparison, the FRM sampler measured a concentration that was 11% higher than the TEOM (50C), 8% higher mass than beta gauge (Metone), 9% higher mass than beta gauge (Anderson), and 2% higher mass than the TEOM (FDMS).

On the basis of the 95% confidence intervals from a one-way ANOVA, no significant difference was found between contemporaneous 24 hr average $PM_{2.5}$ mass concentrations of beta gauge (Anderson) and FRM, or beta gauge (Metone) and FRM, or TEOM (FDMS) and FRM. However, the ANOVA results did show a statistical significance for the measurements made by FRM and TEOM (50C). Thus, most of the continuous monitors had performance comparable to the FRM sampler measurements of $PM_{2.5}$ mass.

A linear regression analysis was performed to examine the relationship between the $PM_{2.5}$ mass concentrations obtained by the FRM and each real-time monitor. Because of the smaller instrument uncertainty of the FRM, it was used as the independent variable and real-time measurements were used as dependent variables. The linear regressions associated with the FRM and each real-time $PM_{2.5}$ monitor during the winter campaign are summarized in Table 4.

The regression between FRM and TEOM (50C) had a positive offset (4.75 ± 0.92 $\mu g/m^3$), which was identified as a manufacturer setting described earlier. The TEOM (50C) underestimated $PM_{2.5}$ mass, as shown in Figure 1, especially when the $PM_{2.5}$ level was greater than 20 $\mu g/m^3$. This was probably due to the loss of semi-volatile mass, which was previously identified during the CEF tests. The low slope (0.54) and regression coefficient (r^2 ~0.81) suggest a large difference existing between performances of TEOM (50C) and FRM sampler during the winter campaign.

The linear regression result found that TEOM (FDMS) reported ambient mass levels that were 17% higher than those measured by the FRM. However, the F-test showed that the mean concentrations measured by the FRM and TEOM (FDMS) were not significantly different during the period. The non-zero intercept (1.6 $\mu g/m^3$) in the regression had affected the regression because TEOM (FDMS) was only used in the period when a small range in $PM_{2.5}$ concentration was encountered at the site (~3–16 $\mu g/m^3$). When the regression intercept was forced to zero, the regression slope improved to 0.98 ± 0.04, indicating that

Table 4. Regression slopes and intercepts for $PM_{2.5}$ mass concentration between paired real-time monitors (dependent variable) and gravimetric samplers (independent variable).

	CEF Study (Summer)				Field Study (Summer)				Field Study (Winter)			
Real-Time Monitor	*N*	B_1	B_0	R^2	*N*	B_1	B_0	R^2	*N*	B_1	B_0	R^2
TEOM (FDMS)	9	1.15	0.44	0.98	27	1.28	−0.90	0.96	9	1.17	−1.57	0.91
TEOM (50C)	9	0.90	0.87	0.93	20	0.95	2.26	0.94	20	0.54	4.75	0.81
Beta gauge (Metone)	9	1.01	1.84	0.98	16	1.15	0.32	0.87	15	0.89	0.50	0.95
Beta gauge (Anderson)	9	1.25	−1.17	0.97			NA		19	0.86	0.40	0.94
SidePak nephelometer	9	3.38	5.81	0.99			NA				NA	
Radiance nephelometer	8	4.90[a]	1.26[a]	0.97			NA				NA	

Notes: [a]Unit E-6 $m^2/\mu g$; *N* = sample size; B_1 = regression slope; B_0 = intercept; R^2 = regression coefficient. HI was used in CEF study and FRM for field study as the gravimetric sampler.

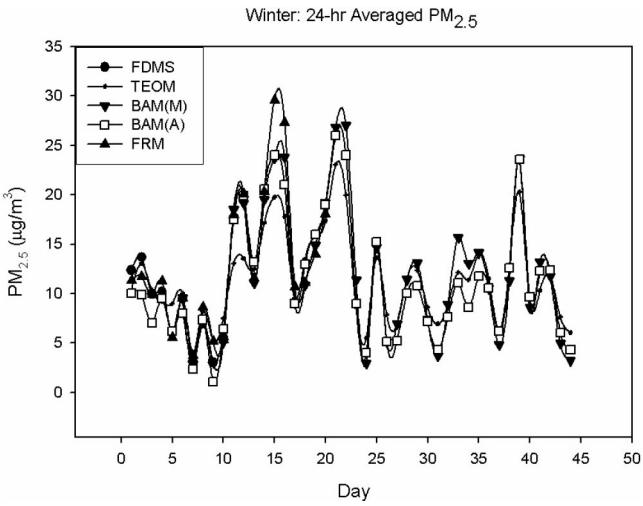


Figure 1. Time series of 24-hr integrated $PM_{2.5}$ measurements made with two beta gauges, two TEOMs, and a FRM sampler at Ewing, NJ, between December 2004 and February 2005.

a regression constrained to zero intercept was more useful in describing the relationship between TEOM (FDMS) and FRM when the $PM_{2.5}$ level was low.

Both the Anderson and Metone beta gauges and the FRM sampler all performed similarly, partly because the temperature of sampling air of these monitors were all heated to 50 °C.

Field Late Summer Study. The response of the real-time monitors and the FRM during September 2004 are illustrated in Figure 2. The hourly $PM_{2.5}$ data from continuous monitors were averaged over 24-hr periods to match the FRM results. A leak was detected in the TEOM (50C) monitor on days 2 and 3; thus, these data were removed in the analysis. Using the same sampling days for comparison, the FRM mass was 15% lower than the beta gauge (Metone) readings, and 14% lower than the TEOM (FDMS) values.

Using the same one-way ANOVA of 95% confidence intervals found no significance difference between mean $PM_{2.5}$ mass concentrations measured by the TEOM (50C) and FRM, or the beta gauge (Metone) and FRM, or the TEOM (FDMS) and FRM. The results indicated that all of the continuous monitors and the FRM sampler performed similarly for measuring the daily variation in $PM_{2.5}$ mass concentration under summer conditions.

A regressions analysis was also performed between the paired FRM and each of the real-time $PM_{2.5}$ monitors, as summarized in Table 4. A positive offset of 2.26 ± 0.69 $\mu g/m^3$ was observed in the regression between FRM and TEOM (50C). This artifact was also associated with the TEOM (50C) during the winter season study. The slope of the regression ($\sim 0.95 \pm 0.06$) indicates that for estimation of $PM_{2.5}$ mass under late summer conditions, the TEOM (50C) is similar to the FRM. The TEOM (FDMS) reported higher particle mass by approximately 28% than the FRM during September 2004 at the field site (Table 4). When compared with the gravimetric method, the beta gauge (Metone) had slightly higher particle mass concentrations.

Effect of Specific Conditions (Temperature and Relative Humidity) on the Performance of Real-Time Monitors

The ambient conditions at the field site associated with the two sampling campaigns were recorded by the collocated weather station. The hourly ambient temperature and RH were approximately −10.7–10.2 °C (mean = 2.1 °C) and approximately 22–98% (mean 68%) during the winter campaign. In September 2004, the ambient temperature and RH were approximately 14.3–23.4 °C (mean 19.7 °C) and approximately 34–96% (mean 81%). These meteorological data were validated using data from a nearby station at Mercer County Airport. The differences of temperature and RH between the two stations were less than 3 °C and 10% RH, respectively.

The difference in $PM_{2.5}$ mass concentration measured between the FRM and each real-time monitor was evaluated as a function of ambient temperature and RH at the field site for the two sampling campaigns. The mass concentration differences are derived by subtracting the mass measured by the continuous methods from the mass measured by the FRM.

In general, there was no correlation among the differences of $PM_{2.5}$ concentrations and the ambient conditions, except for the comparison between the FRM and the TEOM (FDMS), as shown in Figure 3. When the ambient temperature was above 20 °C, the differences in mass concentrations between the TEOM (FDMS) and FRM increased as the ambient temperature increased ($r^2 = 0.32$). The TEOM (FDMS) monitor tended to estimate higher $PM_{2.5}$ mass concentrations than the FRM, especially during warmer weather. This difference in mass concentration is attributed to the loss of semi-volatile mass by the FRM. It is well known that volatilization of particulate nitrate and organic compounds occurs during FRM sampling.[1] On the other hand, the TEOM (FDMS) real-time monitor effectively separates and captures both the volatile and nonvolatile aerosol fraction[12]; thus, the difference between the TEOM (FDMS) and FRM could be explained by the volatilization of semi-volatile species present in the PM at high ambient temperatures. At a

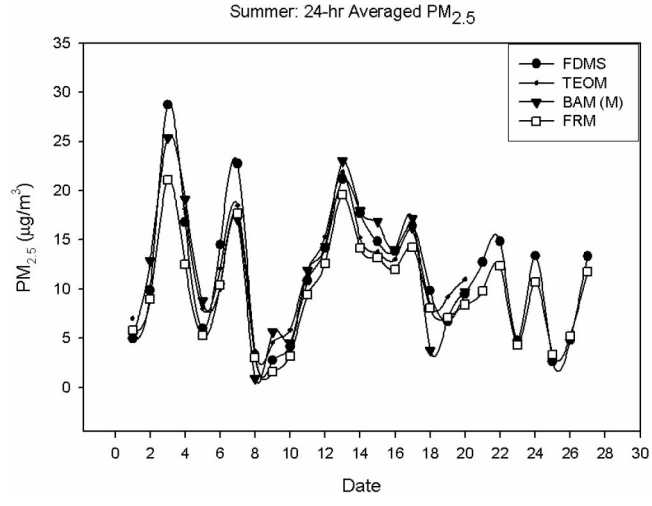


Figure 2. Time series of PM measurement made with two beta gauges, two TEOMs, and a FRM sampler at Ewing, NJ, in September 2004.

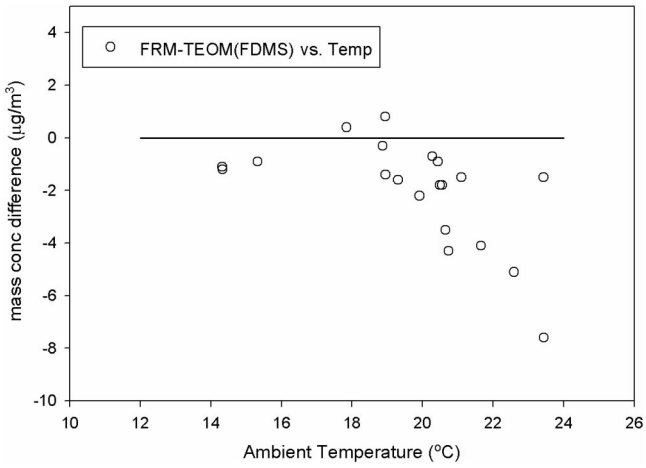


Figure 3. Difference of $PM_{2.5}$ mass concentration measured between the FRM and TEOM (FDMS) monitor as a function of ambient temperature at field site during summer sampling campaign.

higher temperature, ammonium nitrate is easily evaporated into gaseous nitric acid and ammonia;[18] thus, the mass difference between the FRM and TEOM (FDMS) followed the change of the ambient temperature. The result confirmed that $PM_{2.5}$ mass loss from the FRM existed when the ambient temperature was above 20 °C, and that TEOM (FDMS) has the potential to quantify the semi-volatile species even during warm sampling conditions and yield more accurate measurement of ambient $PM_{2.5}$.

RH did not correlate with the difference in $PM_{2.5}$ concentrations obtained by the gravimetric and the real-time measurements in this study. Walter[16] reported that RH changed the size distribution of ambient aerosols; however, for FRM, TEOMs, and beta gauges, the effect of particle size changes on the measured mass was not significant because the filters used by these instruments effectively removed 99% of the suspended particles, regardless of size range or sampling flow rate.[17]

Another influence of RH on mass measurements could be particle-bound and condensed water vapor,[1] which are removed during the RH equilibration of the FRM in the laboratory or during sampling processes of continuous methods using heating devices/diffusion dryer. Low humidity facilitates semi-volatile loss;[18] for example, when the RH is less than the deliquescent point of ammonium nitrate (i.e., 62% at 298 K), it is easy for ammonium nitrate to evaporate. When RH is greater than the deliquescent point, ammonium nitrate exists as a saturated salt solution and stays on filters. In our study, periods with high RH were encountered in both seasons (average RH greater than 62%); thus the mass differences between FRM and continuous methods, those that were seasonal, cannot be attributed directly to changes in RH.

During the field evaluations, the average wind speed was stronger in the winter (average of 6.3 km/hr and range of ~0–21.4 km/hr) than in September (average of 3.2 km/hr and range of ~0–18.2 km/hr). Wind speed has been found to affect the performance of the size-selective inlet,[19] an important component of $PM_{2.5}$ monitors. The size-selective inlet assemblies usually comprise an ambient 10-μm inlet followed by a 2.5-μm inlet (cyclone or impactor) and extended down the tube. The federal registered PM_{10} inlet used in this study satisfied the requirement that the coefficient of variation of three collocated inlets is less than 15% when the wind speed is in the range of 2–24 km/hr.[20] Because the wind speed encountered in the field test stayed in this range, we believe that wind speed did not significantly influence the performance of $PM_{2.5}$ monitors.

Seasonal Differences and Particle Composition

Seasonal differences in the monitor performance were found between paired monitors, and the regression relationships are shown in Figure 4. The slopes of the summer regression were generally higher than those of winter, with the largest difference observed between the FRM and the TEOM (50C) monitor. This seasonal difference was consistent with the findings in the CEF tests, and a study conducted in Boston, MA,[21] which showed that the gravimetric and continuous methods had good agreement in summer and poor agreement in winter. The result could be explained by variations in local meteorological conditions and/or particle composition.

The meteorological conditions associated with the two sampling seasons were described earlier and discussed for their effects on the instrument performance in the previous section (Figure 3). When compared with the FRM, the performance of TEOM (50C) did not depend on the ambient conditions during each campaign, although a wide temperature range (i.e., −10.7~10.2 °C) was encountered in the winter campaign. Therefore, a conclusion could be drawn that when the particle composition is stable, meteorological conditions do not appear to influence the performance of the TEOM (50C) monitor.

Interpretation of Real-Time $PM_{2.5}$ Data during an "Episode"

Transient increases in PM loading may trigger significant health events in sensitive populations, although to date this has not been translated into health-based short-period PM concentration standards.[8] Current air quality standards are based around 24-hr exposure data and associated health effects. Continuous measurements offer

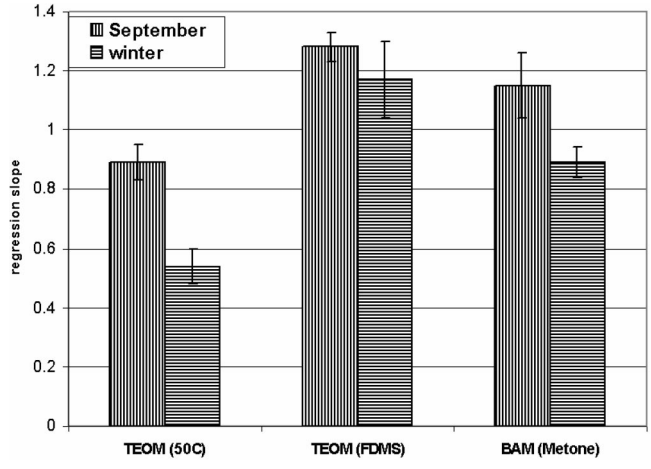


Figure 4. Regression slope between FRM (independent variable) and continuous methods under different seasons at the field site.

the prospect of predicting the likelihood of exceeding a 24-hr standard near real time and thus providing an element of early warning of a likely exceedence; this can also form the basis of an AQI. The method of averaging continuous PM data to give the most informative 24-hr particulate mass loading in this case becomes important and several methods have been proposed. To evaluate the relative effectiveness of some of these methods for reporting $PM_{2.5}$ on a continuous basis, as would be required for an AQI, several methods are compared in Figure 5 for the Ewing, NJ field during a period that should be categorized as "unhealthy for sensitive groups" by the AQI (i.e., $PM_{2.5}$ concentration in the range of ~40.5–65.4 $\mu g/m^3$). Each point in Figure 5 is a rolling average, except the plot of the hourly average method. The hourly $PM_{2.5}$ mass concentrations were determined by the TEOM (FDMS) monitor.

The hourly average method indicated the AQI in the orange zone (i.e., $PM_{2.5}$ concentration in the range of ~40.5–65.4 $\mu g/m^3$) from 8:00 p.m. October 29, 2004 to 4:00 a.m. October 30, 2004. After 4:00 a.m., the PM concentration quickly dropped and the AQI fell to the moderate range (i.e., ~15.5–40.4 $\mu g/m^3$).

The mid-24 hr average method was regarded as the best 24-hr average for the AQI.[22] The mid-24 hr average method had the same response as the hourly average method as shown in Figure 5, but it reported a much lower maximum value of 41.9 $\mu g/m^3$. Therefore, the mid-24 hr average method did not show the orange zone for 6 hr. In addition, calculations by the mid-24 hr average method require the $PM_{2.5}$ concentration in the subsequent 12 hr; thereby, it is not feasible for the purpose of AQI reporting.

The 24-hr end averaged method has comparable performance as the hourly average method when there is no abrupt change of $PM_{2.5}$ level. However, in this episode, the 24- hr end averaged method had the worst performance in this episode among all methods, reporting a maximum value of 9 hr behind that of the hourly average method, making it an ineligible candidate for AQI reporting.

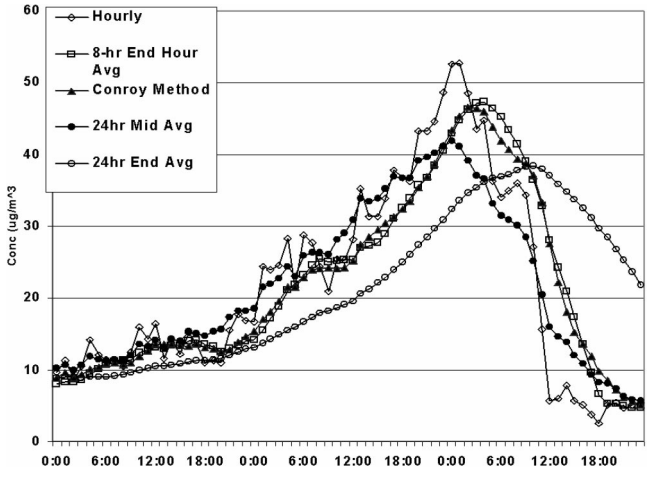


Figure 5. $PM_{2.5}$ data from the TEOM (FDMS) monitor at Ewing, NJ for the period from 12:00 a.m. October 29, 2004 through 11:00 p.m. October 31, 2004. Reporting methods include: hourly average, 8-hr end-hour average, Convoy method, 24-hr mid-hour average, and 24-hr end-hour average.

The Conroy method reported orange zone and maximum value of 46.6 $\mu g/m^3$ 3 hr later than the hourly average method. The 8-hr end-hour average method displayed comparable results as the Conroy method and reported a maximum value of 47.4 $\mu g/m^3$, but 3 hr behind the hourly average method. Therefore, the Conroy method and the 8-hr end-hour average method are reasonable tools for use in the real-time reporting of AQI, especially during an episode. More episodes need to be evaluated before this method can be considered as the best alternative.

CONCLUSIONS

Several time-resolved particle measurement technologies can be used to track particle concentrations in a variety of environments. Most of the mass-based continuous monitors achieve a low detection limit, good reliability, and have a strong correlation with the gravimetric method. Sampling conditions and particle properties can impact the performance of real-time instruments and their same effects can be corrected or eliminated through detailed evaluation before use.

For the purpose of regulatory monitoring, the results obtained at a field site showed that all continuous $PM_{2.5}$ monitors were able to respond to the changes of meteorological conditions when used to provide daily averaged concentrations. Each monitor has the potential for use as a replacement for the FRM. The TEOM (FDMS) monitor, however, provided the most accurate measurement of mass concentration due to minimal mass loss, especially in the warm weather. The TEOM (FDMS) is recommended for use in routine and pollution-episode monitoring, and PM exposure and health studies. Our work supports several previous studies showing that the TEOM (50C) monitor can significantly underestimate the particle mass, especially in the wintertime, and thus should be used with caution in the cold weather. The beta gauges had performance that was comparable to the gravimetric method results under a variety of ambient conditions. The Conroy method and the 8-hr end-hour average method is a reasonable tool for use in the real-time reporting of AQI, especially during an episode.

ACKNOWLEDGMENTS

This research has been supported and funded by various organizations, including the New Jersey Department of Environmental Protection grant (SR03-070). The authors greatly appreciate the advice and suggestions from Dr. Alexander Pollisar and Charles Pietarinen, and technical support from Tom McKenna. Drs. Lioy and Zhang are also supported as part of a National Institute for Environmental Health Sciences grant (ES050-22-12).

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